Predicting Global Crop Production and Analyzing Food Security Using FAOSTAT Data

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ABSTRACT

Global food security is increasingly challenged by population growth, climate variability, resource constraints, and fluctuations in agricultural productivity. Accurate prediction of crop production can support governments, policymakers, agricultural organizations, and researchers in sustainable agricultural planning, resource allocation, and food security initiatives. This capstone project develops a machine learning framework for predicting global crop production and analyzing food security using historical datasets obtained from the Food and Agriculture Organization's FAOSTAT database. Multiple agricultural and socioeconomic datasets were integrated, including crop production, land use, fertilizer use, population statistics, and food security-related information.

The project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology and includes data acquisition, preprocessing, exploratory data analysis, feature engineering, machine learning model development, and model evaluation. Four supervised regression models—Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor—were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. XGBoost achieved the lowest MAE, while Linear Regression achieved the lowest RMSE and highest R^2. However, the relatively low maximum R^2 of 0.0637 indicates that substantial variation in crop production remains unexplained by the current predictors. The resulting framework provides a reproducible foundation for agricultural prediction and is deployed through an interactive Streamlit dashboard that enables users to explore historical agricultural trends, visualize model performance, and generate crop production predictions.

KEYWORDS

Crop Production Forecasting; Food Security; FAOSTAT; Machine Learning; Predictive Analytics; Sustainable Agriculture; CRISP-DM; Streamlit Dashboard.

INTRODUCTION

Agriculture plays a fundamental role in supporting economic development, environmental sustainability, and global food security. It provides livelihoods for billions of people while supplying the food needed to sustain a growing population. However, agricultural production is becoming increasingly difficult to predict due to factors such as climate change, population growth, land degradation, changing weather patterns, and fluctuations in the availability of natural resources. These challenges have made food security a growing global concern and highlighted the need for more reliable approaches to agricultural planning.

The scale of this challenge is reflected in recent international reports. According to the **Food and Agriculture Organization (FAO)**, approximately **733 million people experienced hunger in 2023**, while nearly one in eleven people worldwide continue to face food insecurity (FAO et al., 2024). At the same time, the **United Nations** estimates that the global population will reach approximately **9.7 billion by 2050**, placing additional pressure on agricultural systems to produce more food using limited resources (United Nations, 2022). These trends demonstrate why improving the ability to forecast agricultural production is no longer only an academic problem but also an important practical challenge for governments, policymakers, researchers, and international organizations.

Traditionally, agricultural planning has relied on historical production records and descriptive statistical analyses to understand past trends. Although these approaches provide valuable information, they offer limited support for anticipating future production under changing environmental and socioeconomic conditions. Recent developments in data science and machine learning provide new opportunities to analyze large volumes of agricultural data, identify complex patterns that may not be apparent through conventional methods, and generate predictive insights that support proactive decision-making.

Against this background, this project investigates the following research question:

Can machine learning models developed using historical FAOSTAT data accurately predict global crop production while providing meaningful insights into food security to support evidence-based agricultural decision-making?

To answer this question, the study will integrate multiple datasets available through the **Food and Agriculture Organization's FAOSTAT** platform, including crop production, land use, fertilizer consumption, food balance sheets, population statistics, food security indicators, and climate-related variables. Rather than examining these datasets independently, the project seeks to combine them within a unified analytical framework to better understand how different agricultural, environmental, and socioeconomic factors influence crop production over time.

The outcomes of this research are intended to support a wide range of stakeholders, including government agencies, policymakers, agricultural organizations, researchers, and international organizations such as the FAO. The proposed machine learning framework and interactive Streamlit dashboard will enable users to explore historical agricultural trends, generate crop production forecasts, and examine food security indicators through an accessible decision-support platform. Such insights can contribute to policy development, agricultural resource planning, and long-term food security strategies.

The primary objective of this capstone project is to design and evaluate an end-to-end data science solution for forecasting global crop production while analyzing food security using historical FAOSTAT data. The project follows the **Cross-Industry Standard Process for Data Mining (CRISP-DM)** methodology, beginning with business understanding and data acquisition, followed by data preprocessing, exploratory analysis, feature engineering, predictive modeling, model evaluation, and deployment through a Streamlit-based application. In addition to developing predictive models, the project aims to provide an interactive tool that allows users to visualize agricultural trends and explore model predictions in a practical and intuitive manner.

This study is guided by the hypothesis that combining multiple agricultural, environmental, and socioeconomic indicators available within the FAOSTAT ecosystem will improve crop production forecasting compared with approaches that rely on a limited set of variables. It is further expected that integrating predictive analytics with interactive visualization will provide stakeholders with more meaningful insights for agricultural planning and contribute to data-driven decision-making aligned with the **United Nations Sustainable Development Goal 2 (Zero Hunger)**.

BACKGROUND

Food security has become one of the most significant global challenges of the twenty-first century. Ensuring that sufficient, safe, and nutritious food is available for an increasing global population requires careful planning, efficient resource management, and informed policy decisions. Agricultural production is influenced by a wide range of interconnected factors, including climatic conditions, land availability, fertilizer usage, irrigation practices, trade, population growth, and economic development. Changes in any of these factors can directly or indirectly affect crop production and, consequently, food availability and food security. As these relationships become increasingly complex, governments and international organizations are relying more heavily on data-driven approaches to support agricultural planning and long-term decision-making.

The **Food and Agriculture Organization of the United Nations (FAO)** plays a central role in monitoring global agricultural systems through **FAOSTAT**, one of the world's largest publicly available agricultural databases. FAOSTAT provides historical and current data covering crop production, land use, livestock, fertilizer consumption, trade, food balance sheets, food security indicators, greenhouse gas emissions, and many other agricultural statistics for countries across the world. Because the database is continuously updated and standardized, it has become an important resource for researchers, policymakers, and international organizations studying agricultural production and food systems.

Recent global trends further emphasize the need for predictive agricultural analytics. Climate variability has increased the frequency of droughts, floods, and extreme weather events that affect crop yields across many regions. At the same time, population growth continues to increase the demand for food, while limitations in arable land, water resources, and agricultural inputs create additional pressure on food production systems. According to the **United Nations**, the world's population is expected to approach **9.7 billion by 2050**, significantly increasing future food demand (United Nations, 2022). Similarly, the **FAO's State of Food Security and Nutrition in the World 2024** reports that approximately **733 million people experienced hunger in 2023**, demonstrating that food insecurity remains a persistent global concern despite advances in agricultural technology (FAO et al., 2024). These observations highlight the importance of developing predictive tools that can support proactive rather than reactive agricultural planning.

Although large volumes of agricultural data are now available through platforms such as FAOSTAT, many organizations continue to rely primarily on descriptive reporting and historical trend analysis when making agricultural decisions. While descriptive analytics helps explain what has happened in the past, it offers limited support for anticipating future production under changing environmental and socioeconomic conditions. Advances in machine learning provide opportunities to move beyond descriptive analysis by identifying complex nonlinear relationships among agricultural variables and generating forecasts that support evidence-based planning. However, many existing studies focus on individual crops, specific countries, or a limited number of variables, making it difficult to develop scalable solutions that support global agricultural decision-making.

The problem addressed in this capstone project is **the lack of an integrated, data-driven framework for forecasting global crop production while simultaneously analyzing food security using multiple agricultural and socioeconomic indicators**. This problem should not be confused with the project's research question. The research question investigates whether machine learning models can accurately forecast crop production using historical FAOSTAT data. The underlying problem, however, is broader: decision-makers often lack accessible predictive tools that integrate multiple dimensions of agricultural data into a unified framework capable of supporting strategic planning and policy development. Addressing this gap has practical value for governments, international organizations, agricultural researchers, and policymakers responsible for improving food security and sustainable agricultural development.

To address this problem, the project has both analytical and artifact-oriented objectives. From an analytical perspective, the study aims to integrate multiple FAOSTAT datasets, perform exploratory data analysis, engineer meaningful predictive features, develop and evaluate supervised machine learning models for crop production forecasting, and examine the relationship between agricultural production and food security indicators. Model performance will be assessed using appropriate regression evaluation metrics to identify approaches that provide reliable and interpretable predictions.

In addition to the analytical objectives, the project aims to develop an interactive decision-support artifact in the form of a **Streamlit dashboard**. The dashboard will enable users to explore historical agricultural trends, visualize relationships among key variables, compare model predictions, and examine food security indicators across countries and time periods. Rather than serving only as a predictive model, the final artifact is intended to function as an accessible decision-support tool that assists policymakers, researchers, and agricultural organizations in exploring data-driven insights for agricultural planning and food security assessment.

Figure 1 presents the conceptual workflow proposed for this study. The project follows the **CRISP-DM** framework, beginning with business understanding and data acquisition from FAOSTAT, followed by data preprocessing, exploratory analysis, feature engineering, machine learning model development, evaluation, and deployment through an interactive dashboard. This structured workflow provides a transparent and reproducible approach that supports both technical rigor and practical decision-making.

LITERATURE REVIEW

1. Food security and sustainable agricultural production have become central topics in agricultural research due to increasing global population, climate variability, and pressure on natural resources. Traditional agricultural

planning has primarily relied on historical production statistics and descriptive analyses to monitor crop trends. While these approaches provide valuable insight into past agricultural performance, they are limited in their ability to forecast future production under changing environmental and socioeconomic conditions. Recent advances in machine learning, artificial intelligence, and big data analytics have enabled researchers to develop predictive models capable of identifying complex relationships among agricultural variables and generating more accurate forecasts. Consequently, data-driven decision support systems have become an important area of research for governments, policymakers, and international organizations seeking to improve food security and agricultural sustainability.

2. The Food and Agriculture Organization of the United Nations (FAO) has played a significant role in facilitating agricultural research by providing FAOSTAT, one of the world's largest publicly available agricultural databases. FAOSTAT contains standardized historical information on crop production, land use, fertilizer consumption, trade, food balance sheets, climate indicators, population statistics, greenhouse gas emissions, and food security measures across nearly every country worldwide (Food and Agriculture Organization, 2024). Because of its comprehensive coverage and standardized reporting framework, FAOSTAT has become the primary data source for numerous agricultural forecasting studies.
3. Early agricultural forecasting methods relied heavily on statistical time-series models such as Autoregressive Integrated Moving Average (ARIMA), exponential smoothing, and linear regression. These approaches perform reasonably well when production follows relatively stable historical patterns and relationships among variables remain largely linear. For example, Box and Jenkins (1976) demonstrated that ARIMA models are effective for forecasting univariate time-series data by modeling temporal autocorrelation within historical observations. Similar forecasting techniques have been widely adopted in agricultural economics to estimate crop yields and production trends.
4. Although classical statistical models remain useful, they often struggle to model the highly nonlinear interactions that characterize modern agricultural systems. Crop production is simultaneously influenced by weather conditions, soil quality, irrigation practices, fertilizer application, land availability, technological development, market dynamics, and socioeconomic factors. These complex interactions make it difficult for conventional regression techniques to accurately represent agricultural production under changing environmental conditions (Lobell & Burke, 2010).
5. The emergence of machine learning has significantly expanded the analytical capabilities available for agricultural forecasting. Unlike traditional statistical models, machine learning algorithms can automatically identify nonlinear relationships and high-dimensional interactions without requiring strict assumptions regarding data distributions. Regression-based machine learning techniques have therefore become increasingly popular for predicting agricultural productivity using heterogeneous datasets.
6. Among supervised learning algorithms, Random Forest has become one of the most widely adopted methods for agricultural

prediction. Introduced by Breiman (2001), Random Forest combines multiple decision trees through bootstrap aggregation to reduce variance while improving predictive accuracy. The algorithm is particularly effective when datasets contain numerous correlated variables and nonlinear interactions, characteristics commonly observed in agricultural datasets. Previous studies have demonstrated that Random Forest provides robust performance for crop yield prediction while remaining relatively resistant to overfitting compared with individual decision trees (Jeong et al., 2016).

7. Gradient boosting methods have also gained widespread attention in agricultural analytics due to their ability to sequentially improve model performance by correcting prediction errors from previous iterations. Algorithms such as Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and CatBoost have consistently achieved strong predictive performance across many regression applications involving structured tabular data (Chen & Guestrin, 2016). Their ability to capture nonlinear relationships while efficiently handling missing values and high-dimensional datasets makes them particularly well suited for large agricultural databases such as FAOSTAT.
8. Support Vector Regression (SVR) has likewise been successfully applied to agricultural prediction problems. SVR extends the principles of Support Vector Machines to continuous prediction tasks by constructing regression functions that maximize predictive accuracy while minimizing model complexity (Drucker et al., 1997). Researchers have demonstrated that SVR performs particularly well when agricultural datasets contain nonlinear relationships and relatively limited sample sizes. However, its computational requirements increase substantially as dataset size grows, making tree-based ensemble methods more practical for global agricultural databases.
9. Artificial Neural Networks (ANNs) have also become widely used in agricultural forecasting due to their ability to approximate highly complex nonlinear functions. Haykin (2009) demonstrated that multilayer neural networks can effectively model intricate relationships among environmental and socioeconomic variables. Numerous agricultural studies have applied neural networks for crop yield prediction, rainfall estimation, irrigation optimization, and climate impact assessment. Nevertheless, neural networks generally require larger training datasets and greater computational resources than tree-based machine learning algorithms while offering reduced interpretability for decision makers.
10. More recently, deep learning architectures such as Long Short-Term Memory (LSTM) networks have shown promising performance for agricultural forecasting involving temporal sequences. LSTM models extend recurrent neural networks by introducing memory cells capable of capturing long-term temporal dependencies within sequential observations (Hochreiter & Schmidhuber, 1997). Several recent studies have applied LSTM models to crop yield forecasting using weather observations, satellite imagery, and historical production data, demonstrating improved performance over traditional statistical models for long-term prediction tasks. Despite these advantages, LSTM models typically require large quantities of high-frequency temporal observations and

extensive computational resources, making their implementation more complex than conventional machine learning methods.

11. Beyond predictive modeling, researchers have increasingly emphasized the integration of multiple agricultural datasets to improve forecasting accuracy. Rather than relying solely on historical crop production records, contemporary studies combine climatic variables, fertilizer application, land utilization, irrigation statistics, food balance information, and demographic indicators to develop more comprehensive predictive models. Such integrated approaches recognize that agricultural production is influenced by numerous interconnected environmental and socioeconomic processes rather than individual variables in isolation.
12. Feature engineering has therefore become a critical component of agricultural machine learning workflows. Previous research has demonstrated that deriving additional explanatory variables—including lagged production values, production growth rates, fertilizer efficiency measures, land productivity ratios, and regional aggregation indicators—can substantially improve predictive performance (Kamilaris & Prenafeta-Boldú, 2018). Proper feature engineering allows machine learning algorithms to better capture long-term agricultural trends while reducing the effects of noisy or redundant information.
13. In addition to predictive accuracy, interpretability has emerged as an important consideration for agricultural decision-support systems. Governments and policymakers require transparent explanations of model predictions before incorporating them into planning decisions. Ensemble tree methods provide several mechanisms for interpreting prediction behavior, including feature importance rankings, partial dependence plots, and SHAP (SHapley Additive exPlanations) values. Lundberg and Lee (2017) demonstrated that SHAP provides theoretically consistent explanations of machine learning predictions, enabling users to understand how individual variables influence model outputs. Explainable artificial intelligence has consequently become an increasingly valuable component of agricultural analytics.
14. While predictive models provide valuable forecasts, many researchers argue that interactive visualization systems substantially improve the practical usability of machine learning solutions. Dashboards developed using modern visualization frameworks allow users to explore historical trends, compare model predictions, evaluate uncertainty, and investigate regional agricultural performance without requiring advanced technical expertise. Streamlit has become one of the most widely adopted frameworks for rapidly deploying Python-based machine learning applications because it integrates data visualization, predictive modeling, and interactive user interfaces within a single development environment (Streamlit, 2024). Recent agricultural decision-support systems have successfully employed Streamlit dashboards to improve accessibility for policymakers and agricultural researchers.
15. Most existing agricultural forecasting studies focus on a single country, a limited geographic region, or individual crop types such as wheat, maize, rice, or soybean. Although these studies demonstrate the effectiveness of machine learning for localized prediction, relatively few investigations integrate multiple FAOSTAT

datasets to construct a comprehensive global forecasting framework that simultaneously analyzes crop production and food security indicators. Furthermore, many existing studies prioritize predictive accuracy while giving comparatively less attention to developing accessible decision-support tools capable of supporting real-world policy decisions.

16. The present capstone project addresses these limitations by integrating multiple FAOSTAT datasets within a unified machine learning framework following the CRISP-DM methodology. Rather than relying exclusively on historical crop production records, the proposed approach incorporates agricultural, environmental, and socioeconomic variables to improve forecasting performance while simultaneously analyzing food security indicators. Multiple supervised regression algorithms will be evaluated to determine their suitability for large-scale agricultural prediction, and the resulting models will be deployed through an interactive Streamlit dashboard that enables users to visualize historical trends, generate production forecasts, and explore food security patterns across countries. By combining predictive analytics with an accessible decision-support artifact, the project contributes to ongoing research at the intersection of machine learning, sustainable agriculture, and evidence-based policymaking while supporting progress toward the United Nations Sustainable Development Goal 2: Zero Hunger.

DATA UNDERSTANDING

Data Summary and Exploratory Data Analysis

This project uses publicly available datasets from the Food and Agriculture Organization's FAOSTAT database, one of the world's most comprehensive repositories of agricultural statistics (FAO, 2024). Multiple datasets were integrated to provide a holistic view of agricultural production and food security. These include crop production, harvested area, fertilizer consumption, land use, food balance sheets, population statistics, and selected food security indicators. Together, these datasets capture agricultural, environmental, and socioeconomic factors that influence crop production across countries and years.

The primary response variable is annual crop production, while explanatory variables include cultivated land area, fertilizer use, population, food availability measures, and other agricultural indicators. These variables were selected because previous research has shown they contribute significantly to agricultural productivity and food security.

Before analysis, the datasets underwent preprocessing to ensure consistency across countries and reporting years. Missing values were handled using appropriate imputation techniques or by removing records with excessive missing information. Duplicate records were eliminated, and variable names, country identifiers, and measurement units were standardized to facilitate dataset integration. Numerical variables were converted to consistent formats, while categorical variables such as country names and crop types were encoded for modeling.

The integrated dataset was assessed for common data quality issues including missing values, duplicate records, inconsistent data types, and merge consistency across multiple FAOSTAT datasets. Variables with excessive missing values, particularly food security indicators, were excluded from the modeling dataset because reliable imputation could introduce bias. Duplicate observations were checked and removed where applicable, while numerical variables were standardized to ensure consistency for downstream analysis.

Target leakage was explicitly evaluated during feature engineering. Although the feature Production per Capita was created to better understand agricultural productivity relative to population, it was derived directly from the target variable (Production). To prevent leakage and ensure fair model evaluation, this feature was removed prior to model training. This approach ensured that the predictive models only learned from independent explanatory variables.

Exploratory Data Analysis (EDA) was conducted to understand the structure and characteristics of the integrated dataset. Summary statistics were calculated to examine the distributions of numerical variables, while histograms and boxplots identified skewness and potential outliers. Correlation analysis was performed to evaluate relationships among predictor variables and the target variable. Scatter plots and time-series visualizations were also used to investigate production trends across countries and identify long-term patterns.

Exploratory data analysis revealed that crop production exhibited a positively skewed distribution, motivating the application of a logarithmic transformation for exploratory purposes. Correlation analysis indicated that agricultural land area and fertilizer application variables showed positive relationships with crop production, supporting their inclusion as predictor variables. Boxplots also revealed several extreme production values; however, these observations represented genuine differences between countries and crops rather than data errors and were therefore retained.

The exploratory analysis revealed noticeable differences in production levels across regions and crops, reflecting variations in agricultural practices, climate, and resource availability. Several predictor variables showed moderate to strong correlations with crop production, suggesting their suitability for predictive modeling. Although some multicollinearity existed among agricultural indicators, tree-based machine learning algorithms are generally robust to correlated features and were therefore considered appropriate for this study.

METHODOLOGY

Project Pipeline

This project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, which provides a structured approach for developing machine learning solutions. The workflow consists of business understanding, data collection, preprocessing, exploratory analysis, feature engineering, model development, model evaluation, and best model selection. The deployment of the selected model through an interactive Streamlit dashboard will be completed in a subsequent phase of the project. The project pipeline consists of six major stages: data collection from multiple FAOSTAT datasets, data integration, exploratory data analysis, data preparation and feature engineering, machine learning model development, and comparative evaluation. This workflow follows the CRISP-DM framework and provides a structured process from raw agricultural data to predictive modeling.

Figure 1 illustrates the overall project pipeline.

Data Preparation and Feature Engineering

Data preparation included merging multiple FAOSTAT datasets using common identifiers, correcting inconsistent data types, removing duplicate records, and excluding variables with excessive missing values. Features not suitable for modeling, such as identifiers and target-derived variables, were removed prior to model training. These preprocessing steps improved data consistency while reducing the likelihood of biased model predictions.

Several domain-specific features were engineered, including Nitrogen per Land, Phosphate per Land, Potash per Land, Arable Land Ratio, Food Supply Gap, Production per Capita, and Log Production. Production per Capita was excluded from model training to prevent target leakage, while Food Supply Gap was removed from the final modeling dataset because of missing values. Tree-based algorithms (Decision Tree, Random Forest, and XGBoost) were trained using the original feature values because these models are insensitive to feature scaling. Standardization using StandardScaler was applied only to Linear Regression to ensure comparable feature magnitudes and stable coefficient estimation.

Modeling

Crop production forecasting was formulated as a supervised regression problem because the target variable, annual crop production, is continuous. The objective was to estimate crop production from the agricultural and socioeconomic predictors retained after data preparation and feature engineering.

The prepared modeling dataset contained 1,598,628 observations and 12 predictor features. The dataset was divided into training and testing subsets using an 80:20 split with random_state=42, resulting in 1,278,902 training observations and 319,726 test observations. Numerical features used by Linear Regression were standardized using a StandardScaler fitted exclusively on the training data and subsequently applied to the test data, preventing information from the held-out test set from influencing preprocessing. Features identified as potential sources of target leakage, including Production per Capita, were excluded before model training.

Four regression algorithms were evaluated: Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor. Linear Regression served as the baseline because it provides a simple and interpretable benchmark for assessing whether more flexible nonlinear methods improve predictive performance. Decision Tree Regression was included as an interpretable nonlinear model, while Random Forest and XGBoost were evaluated as ensemble approaches capable of capturing more complex relationships and interactions among agricultural predictors.

All four models were trained using the same modeling dataset and evaluated on the same held-out test set to provide a consistent comparison. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The final model was selected based on comparative predictive performance, model interpretability, computational efficiency, and suitability for deployment within the Streamlit application.

Model Evaluation

Model performance was evaluated using three complementary regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). MAE measures the average absolute difference between observed and predicted production values and therefore provides an intuitive measure of typical prediction error. RMSE gives greater weight to large prediction errors and was used to assess the models' sensitivity to substantial deviations between predicted and observed production. R^2 measures the proportion of variation in crop production explained by the predictors included in the model.

The metrics were evaluated together rather than relying on a single measure of performance. Lower MAE and RMSE values indicate smaller prediction errors, whereas a higher R^2 indicates greater explanatory performance. The four models were compared using the same held-out test observations to ensure that differences in performance reflected model behavior under a consistent evaluation setting.

Project Artifact

The analytical outputs generated by the selected Linear Regression model have been integrated into an interactive Streamlit dashboard that serves as the primary decision-support artifact for this project. The application enables users to explore historical crop production trends, visualize agricultural indicators, compare model performance, and generate crop production predictions for selected countries and years.

The dashboard consists of multiple interactive pages, including an overview dashboard, exploratory data analysis visualizations, crop production prediction, food security analysis, and model evaluation results. Users can interactively filter countries, crops, and years while exploring the integrated FAOSTAT datasets through dynamic charts and tables.

The Streamlit application provides an accessible interface for policymakers, researchers, agricultural organizations, and students who wish to analyze historical agricultural data and evaluate machine learning predictions without interacting directly with the underlying Python code.

The complete implementation and source code are available in the project's GitHub repository:

https://github.com/anishchoudhury464/Predicting-Global-Crop-Production-and-Analyzing-Food-Security-Using-FAOSTAT-Data

Deployment

The trained Linear Regression model was serialized using Joblib and integrated into a Streamlit application. The application loads the saved preprocessing pipeline and trained model to generate predictions from user-selected inputs. Visualizations were implemented using Plotly to provide interactive exploration of historical agricultural trends and model outputs. This deployment demonstrates the complete machine learning lifecycle from data acquisition and model development to end-user delivery.

RESULTS

Model Performance Comparison

The four regression models were evaluated on the held-out test dataset using MAE, RMSE, and R^2. Table 2 summarizes the comparative performance of Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor. The results demonstrate differences in predictive performance across the models and provide a quantitative basis for identifying the strongest modeling approach under the current experimental design.

Interpretation of Results

The comparative evaluation revealed that increasing model complexity did not result in substantial improvements over the Linear Regression baseline. XGBoost achieved the lowest MAE, approximately 4.03 million, compared with approximately 4.08 million for Linear Regression. However, Linear Regression produced the lowest RMSE and highest R^2 among the four models. Decision Tree and Random Forest produced similar results, with R^2 values of approximately 0.0574, while XGBoost achieved an R^2 of 0.0586.

The relatively small differences among the four models suggest that the more complex tree-based methods did not provide a meaningful overall advantage over the simpler linear baseline for the current feature set. This finding emphasizes the importance of comparing complex machine learning algorithms against interpretable baseline models rather than assuming that greater model complexity will necessarily produce superior predictive performance.

The highest observed R^2 was 0.0637, indicating that the current predictors explain only a limited proportion of the variation in crop production within the held-out test data. This suggests that substantial variation in global crop production is associated with factors not represented in the current modeling features, potentially including weather conditions, soil characteristics, irrigation, extreme events, crop-specific characteristics, and agricultural management practices. Therefore, the models should be interpreted as exploratory predictive tools rather than highly accurate production forecasting systems.

Discussion of Results

The evaluation results indicate that all four regression models were able to learn meaningful relationships from the integrated FAOSTAT dataset, although overall predictive performance remained modest. Linear Regression achieved the highest coefficient of determination ($R^2 = 0.0637$) and the lowest RMSE, while XGBoost produced the lowest MAE. These findings suggest that although more complex ensemble methods slightly reduced average prediction error, they did not substantially improve the proportion of variance explained compared with the simpler Linear Regression model. Therefore, Linear Regression was selected as the final deployment model because it achieved the highest R^2 score, the lowest RMSE, provided greater interpretability than ensemble methods, required minimal computational resources, and was straightforward to integrate into the Streamlit application.

From a practical perspective, the relatively low R^2 values indicate that the current predictor variables capture only a small portion of the factors influencing global crop production. This outcome is expected because agricultural productivity is affected by many additional variables including rainfall, temperature, soil quality, irrigation practices, crop genetics, and extreme weather events that were not available within the current integrated dataset.

Prediction Performance

Figure 2. Actual versus predicted crop production for the selected model.

The actual-versus-predicted comparison provides a visual assessment of model performance across the test observations. Predictions located close to the diagonal reference line represent observations for which predicted production closely matches actual production, while greater deviations from the line indicate larger prediction errors. The observed dispersion illustrates that the model captures part of the variation in crop production but also experiences difficulty predicting some observations, particularly where production values differ substantially across countries, crops, and years.

Feature Importance Analysis

Feature importance from the XGBoost model provided additional insight into the predictors contributing to crop production predictions. Population was the most influential feature, with an importance score of 0.2961, followed by Phosphate (0.2233), Nitrogen (0.1238), Agricultural Land (0.1112), and Cropland (0.0819). Together, these results suggest that population, fertilizer inputs, and agricultural land availability contain substantial predictive information within the fitted XGBoost model.

The dominance of Population and fertilizer-related variables also supports the rationale for integrating agricultural and socioeconomic information within the modeling framework. However, feature importance represents the contribution of variables to predictions within the fitted model and should not be interpreted as evidence that these variables causally determine crop production.

Although Population, Phosphate, Nitrogen, Agricultural Land, and Cropland were identified as the most influential predictors in the XGBoost model, feature importance should not be interpreted as evidence of causality. Instead, these variables represent those that contributed most strongly to the model's predictions under the current dataset. The findings nevertheless support the project's objective of integrating agricultural and socioeconomic indicators to improve predictive modeling.

Evaluation of the Working Hypothesis

The project hypothesized that integrating multiple agricultural and socioeconomic indicators from FAOSTAT could support improved crop production forecasting. The modeling results demonstrate that the integrated dataset can be used to construct predictive regression models, and the feature-importance analysis indicates that multiple agricultural and socioeconomic predictors contribute information to the modeling process.

However, the strongest model achieved an R^2 of 0.0637, indicating that the current feature set explains only a limited proportion of variation in crop production. Furthermore, the present experiment compares different algorithms using the same integrated feature set rather than directly comparing the integrated dataset against a reduced-feature baseline. Therefore, the results support the feasibility of the integrated modeling framework but do not conclusively demonstrate that data integration itself improves forecasting accuracy. Additional feature-ablation experiments and incorporation of environmental predictors would be required to evaluate that hypothesis more comprehensively.

Practical Implications

The developed modeling framework provides a reproducible workflow for analyzing agricultural production using integrated FAOSTAT datasets. Although prediction accuracy remains moderate, the framework enables governments, policymakers, agricultural researchers, and international organizations to compare historical production patterns, examine the influence of agricultural inputs, and generate data-driven production estimates. Rather than replacing expert judgment, the predictive models are intended to complement existing agricultural planning processes by providing additional analytical evidence for decision making.

The framework is intended to complement existing agricultural planning processes rather than replace expert decision-making. The generated predictions should therefore be interpreted alongside domain expertise, weather information, and other agricultural indicators.

Limitations and Practical Implications

Several limitations should be considered when interpreting the results. FAOSTAT combines observations reported across many countries, crops, and years, which introduces substantial heterogeneity into the modeling problem. Missing observations and differences in reporting practices may also affect data completeness and consistency. In addition, the current feature set does not incorporate high-resolution weather, soil, irrigation, satellite, or crop-management information that may explain additional variation in production.

Despite these limitations, the modeling framework provides a reproducible basis for comparing crop production prediction methods using integrated agricultural data. The results can support exploratory agricultural planning by identifying relationships within historical data and generating model-based estimates. However, predictions should be interpreted alongside domain expertise and other sources of agricultural information rather than used as the sole basis for high-stakes policy or resource-allocation decisions.

Actionability of the Project

The results of this project provide stakeholders with a practical analytical framework for exploring historical agricultural production and generating predictive estimates using machine learning models. The Streamlit dashboard developed as the deployment artifact will enable users to visualize agricultural indicators, compare historical trends, generate production forecasts, and examine model outputs through an interactive interface. These capabilities allow users to perform exploratory analyses more efficiently than relying solely on static reports or manually examining historical datasets.

CONCLUSION

This study developed an end-to-end machine learning framework for forecasting global crop production using integrated FAOSTAT datasets. The project successfully combined agricultural production, land use, fertilizer consumption, and population statistics into a unified analytical pipeline capable of generating exploratory crop production predictions and supporting agricultural trend analysis.

Comparative evaluation demonstrated that Linear Regression achieved the strongest overall performance under the current experimental design by obtaining the lowest RMSE and the highest coefficient of determination (R^2), while XGBoost achieved the lowest Mean Absolute Error. These findings indicate that increasing model complexity did not necessarily improve predictive performance for the available feature set.

The project hypothesis was partially supported. Integrating multiple agricultural and socioeconomic variables enabled the construction of predictive machine learning models capable of estimating crop production. However, the relatively low R^2 values demonstrate that important explanatory variables remain absent from the current dataset. Additional environmental and crop-specific information will likely be required to substantially improve forecasting accuracy.

Overall, this project provides a reproducible end-to-end machine learning framework that supports agricultural analytics and has been successfully deployed as an interactive Streamlit decision-support application for exploring FAOSTAT data and generating crop production predictions.

RECOMMENDATIONS

Several opportunities exist to further improve the predictive capability and practical usefulness of this project. The highest priority should be expanding the available datasets by incorporating additional environmental variables such as rainfall, temperature, soil quality, drought indices, irrigation statistics, and satellite imagery. These variables would allow the models to capture environmental influences that are not represented within the current FAOSTAT datasets.

Future work should also investigate more advanced machine learning and deep learning approaches, including LightGBM, CatBoost, Long Short-Term Memory (LSTM) networks, and Transformer-based forecasting models. These methods may further improve predictive performance by capturing complex temporal relationships within agricultural production data.

From a deployment perspective, the Streamlit dashboard could be enhanced by supporting real-time FAOSTAT data updates, downloadable prediction reports, interactive geographical visualizations, and country-specific forecasting dashboards. Incorporating Explainable AI techniques such as SHAP values would further improve transparency by helping users understand the contribution of each feature to individual predictions.

Finally, future development should focus on deploying the system as a continuously monitored production application that periodically retrains the predictive models as new FAOSTAT data become available. Such an automated workflow would improve long-term prediction accuracy while ensuring that the system remains relevant as global agricultural conditions evolve.

Future work should focus on enhancing the deployed Streamlit dashboard by integrating real-time FAOSTAT data updates, automated model retraining, explainable AI techniques (SHAP), and cloud deployment for broader accessibility.

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APPENDICES

Appendix A. FAOSTAT Datasets Used

The project integrates multiple datasets obtained from the Food and Agriculture Organization (FAO) through the FAOSTAT platform. Table A1 summarizes the primary datasets used during analysis.

Table A1. Variables Summary

Variable	**Type**	**Role**
Area	Identifier	Country
Item	Identifier	Crop
Year	Predictor	Time
Population	Predictor	Population size
Agricultural Land	Predictor	Available agricultural land
Cropland	Predictor	Cultivated land
Nitrogen	Predictor	Fertilizer input
Phosphate	Predictor	Fertilizer input
Potash	Predictor	Fertilizer input
Production	Target	Annual crop production

Table A2. FAOSTAT Datasets Used

Dataset	Description	Purpose
Crop Production	Annual production quantities for major crops	Target variable for prediction
Land Use	Agricultural land and harvested area	Measure available cultivation resources
Fertilizer Use	Annual fertilizer consumption	Predictor of agricultural productivity
Food Balance Sheets	Food availability and consumption statistics	Food security analysis
Population Statistics	Country population estimates	Demand-side indicator
Food Security Indicators	Indicators related to undernourishment and food availability	Food security assessment

Appendix B. Feature Engineering

Several derived variables were created to improve predictive performance.

Table B1. Engineered Features

Feature	Description
Production_per_Capita	Crop production divided by population, representing production available per person.
Nitrogen_per_Land	Nitrogen fertilizer application normalized by agricultural land area to measure nitrogen intensity.
Phosphate_per_Land	Phosphate fertilizer application normalized by agricultural land area to measure phosphate intensity.
Potash_per_Land	Potash fertilizer application normalized by agricultural land area to measure potash intensity.
Arable_Land_Ratio	Ratio of arable land to total agricultural land, representing the proportion of cultivable land.
Food_Supply_Gap	Difference between food supply (kcal/capita/day) and dietary energy requirement, indicating surplus or deficit in food availability.

Feature	**Description**
Log_Production	Natural logarithm transformation of crop production (log1p) used to reduce skewness in the production distribution.

Note: Production_per_Capita was excluded from model training to prevent target leakage. Food_Supply_Gap was removed from the final modeling dataset because the underlying variables contained substantial missing value

Table 2: Model Performance Comparison

Model	MAE	RMSE	R^2
Linear Regression	4,084,985.80	28,860,372.08	0.0637
Decision Tree Regressor	4,044,786.56	28,957,427.83	0.0574
Random Forest Regressor	4,045,803.82	28,957,198.08	0.0574
XGBoost Regressor	4,032,441.51	28,938,300.19	0.0586

Figure 1: Project Pipeline Flowchart

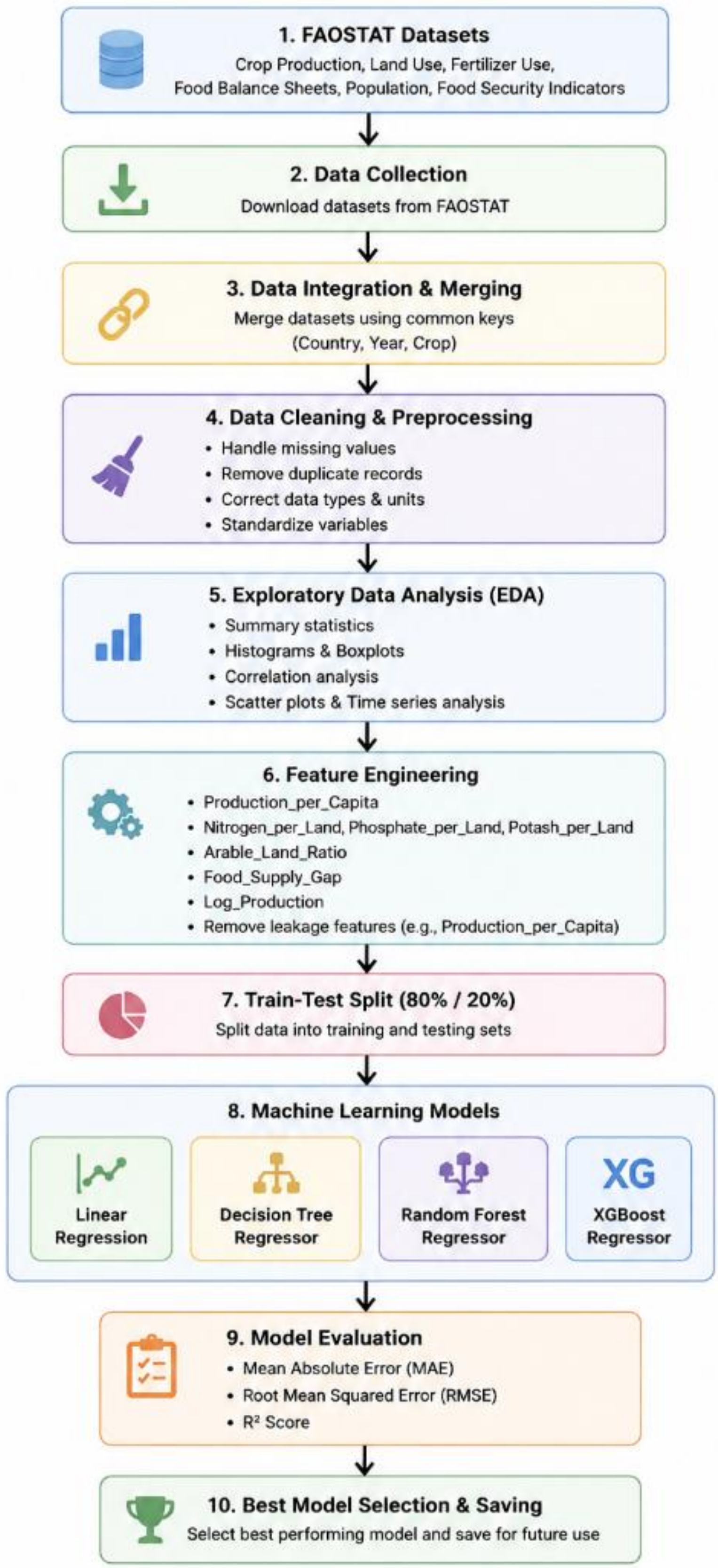


Figure 2: Linear Regression: Actual vs Predicted

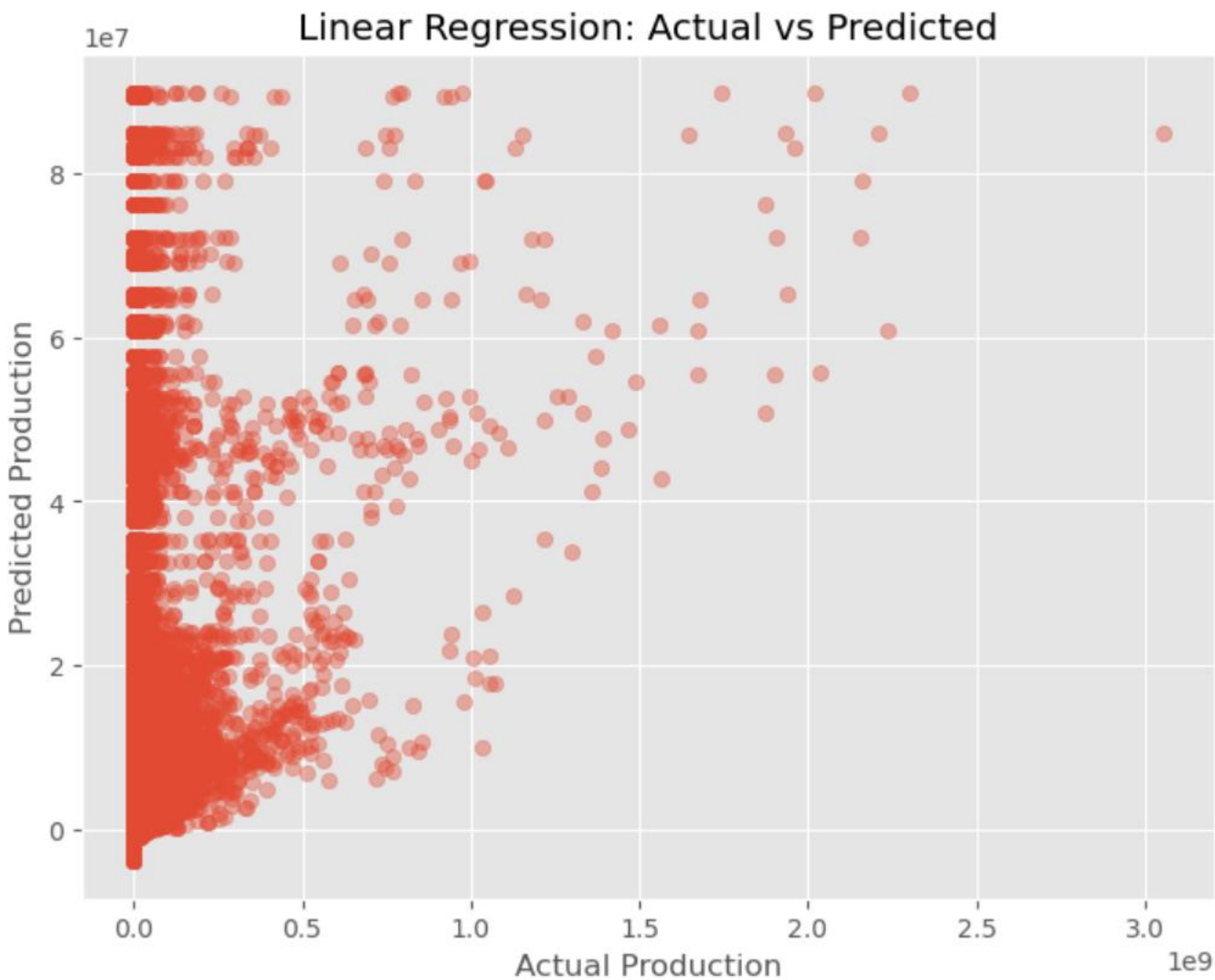
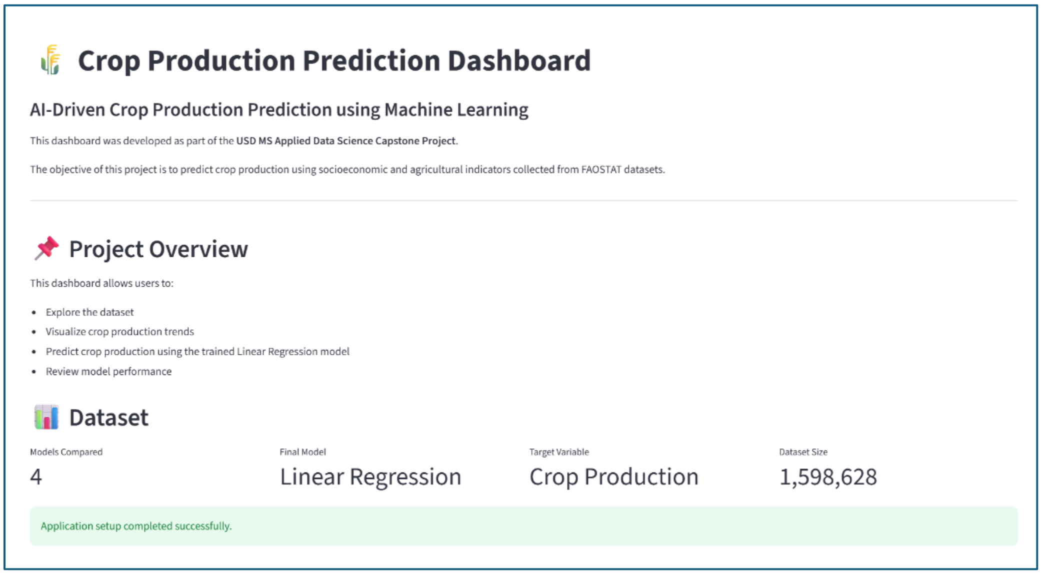
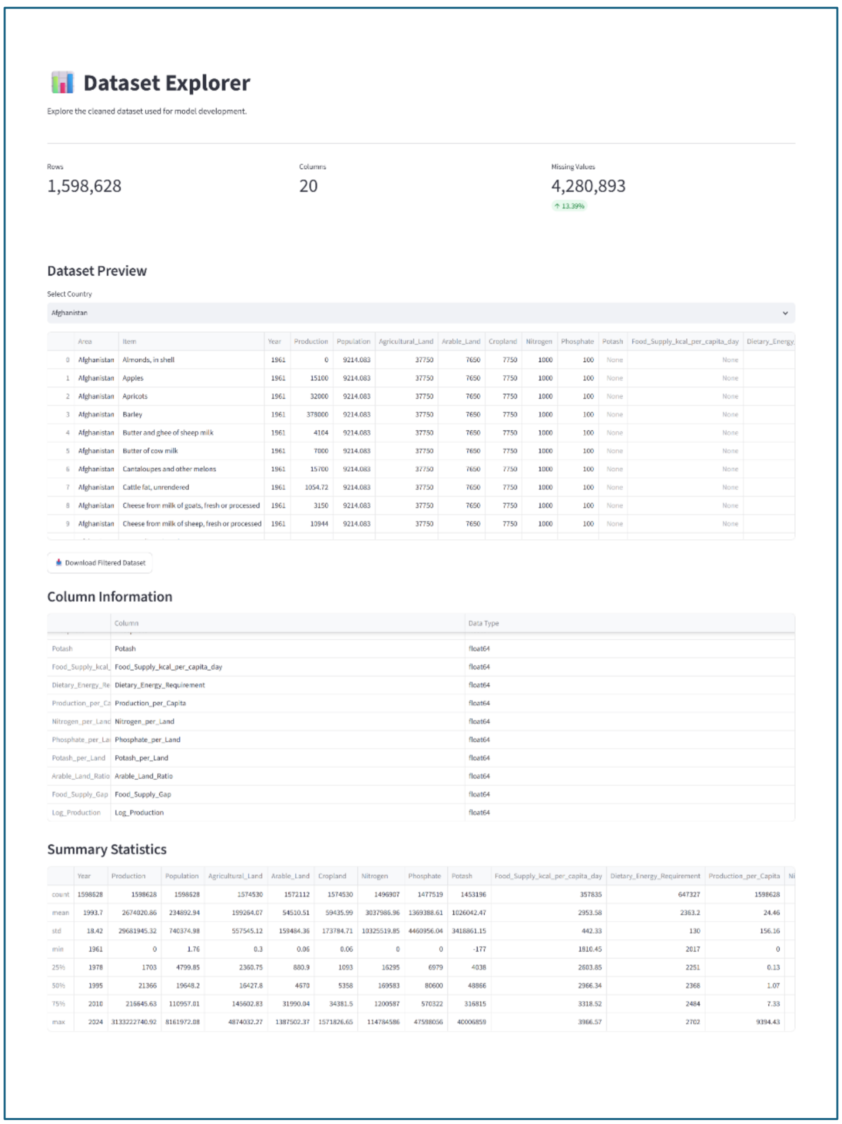
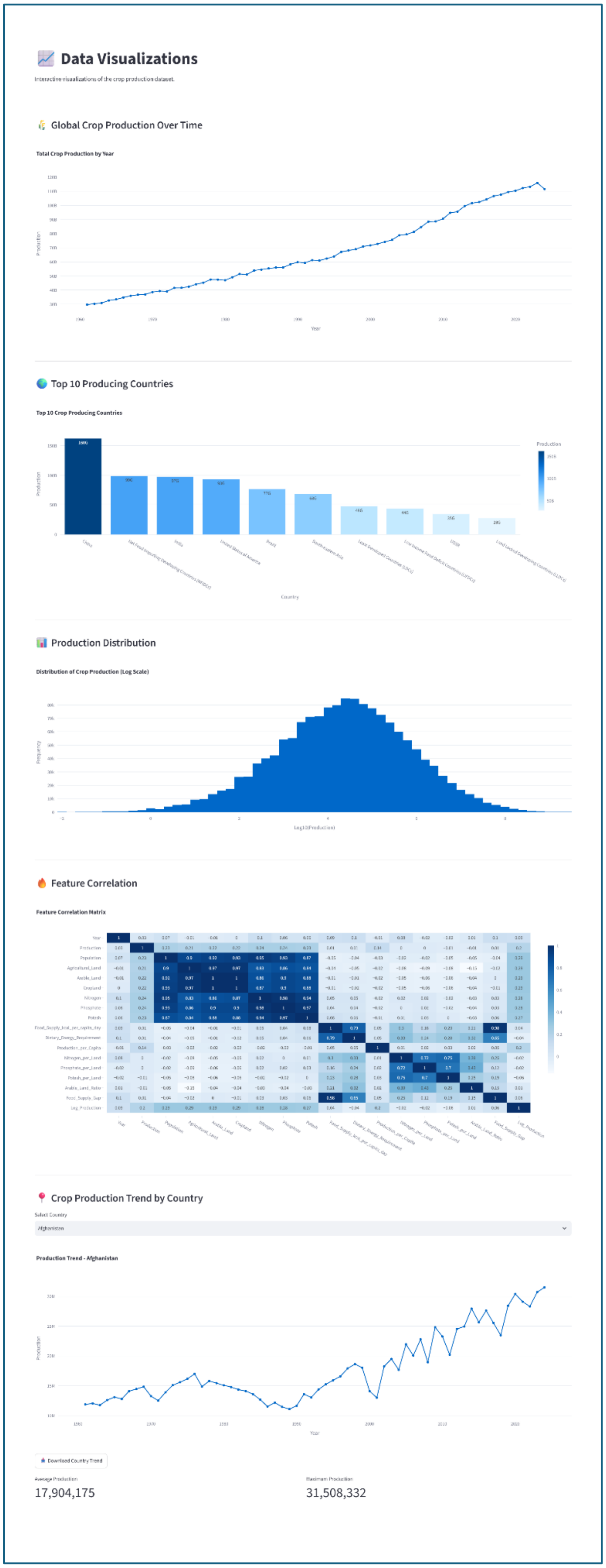


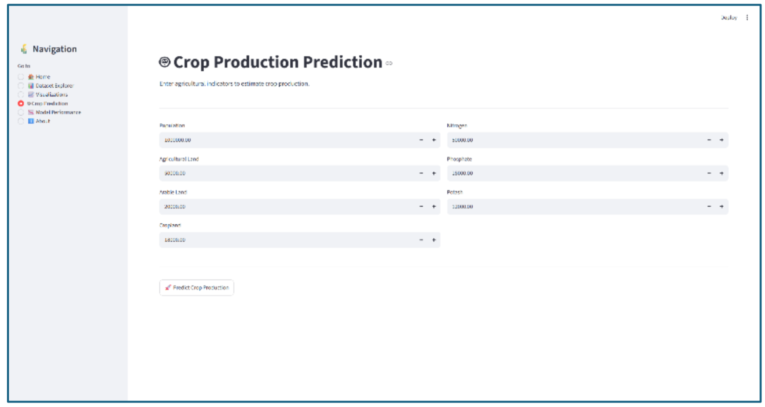
Figure 3. Streamlit Dashboard



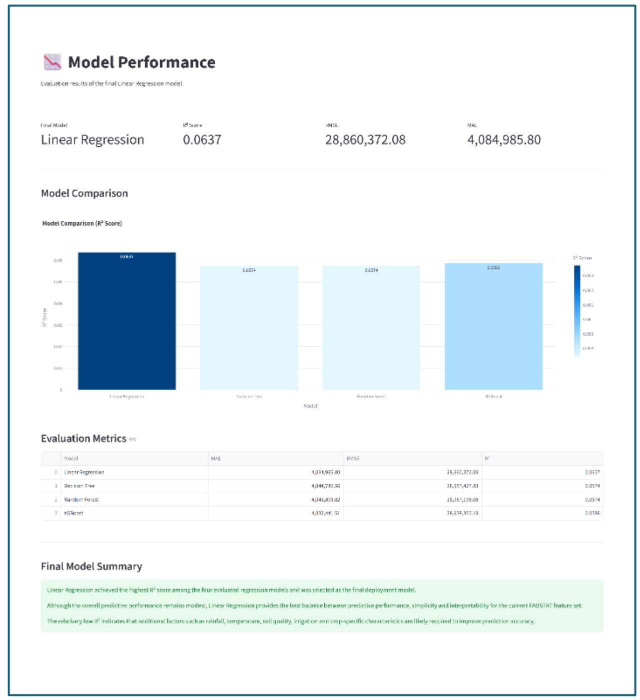




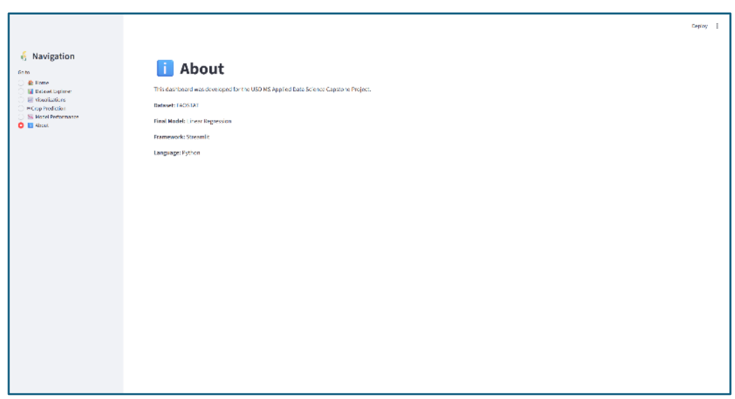
Data Visualizations
Interactive visualizations of the crop production dataset.
Global Crop Production Over Time
Total Crop Production by Year
Top 10 Producing Countries
Top 10 Crop Producing Countries
Production Distribution
Distribution of Crop Production (Log Scale)
Feature Correlation
Feature Correlation Matrix
Crop Production Trend by Country
Select Country
Afghanistan
Production Trend - Afghanistan
Download Country Trend
Average Production
17,904,175
Maximum Production
31,508,332



Navigation
Crop Production Prediction
Predict Crop Production



Model Performance
Linear Regression
0.0637
28,860,372.08
4,084,985.80
Model Comparison
Model Comparison (R² Score)
Evaluation Metrics
Final Model Summary



Navigation
About